

# Lecture 1: Mathematical Foundations for Lattice-Based Cryptography

Groups, Rings, Fields, Modular Arithmetic,  
Vector Spaces, Basis, and Lattices

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Lecture Notes — Session 1 (approx. 2 hours)

**Purpose of this lecture.** Every technical term used later in the thesis proposal — “lattice,” “modulus,” “ring  $\mathbb{Z}_q[x]/(x^n + 1)$ ” — rests on a small set of algebraic building blocks. This lecture builds those blocks, one at a time, from the ground up: *Group*  $\rightarrow$  *Subgroup*  $\rightarrow$  *Ring*  $\rightarrow$  *Field*  $\rightarrow$  *Modular Arithmetic*  $\rightarrow$  *Vector Space*  $\rightarrow$  *Basis*  $\rightarrow$  *Lattice*. Nothing here requires prior abstract algebra; every definition is followed immediately by concrete, worked examples.

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# 1 Why Start With Algebra?

Lattice-based cryptography is, at its core, about doing arithmetic on *vectors* whose coordinates live in a *finite, wraparound number system*, arranged according to strict algebraic rules. Before we can even state what a lattice is, we need precise answers to questions like:

- What does it mean for a set of numbers to have a well-behaved notion of “add” and “multiply”? (**Groups, Rings, Fields**)
- What happens when arithmetic “wraps around,” like a clock? (**Modulus**)
- What does it mean to build every point in a space out of a small number of building-block directions? (**Vector spaces, Basis**)
- What happens when we only allow *whole-number* combinations of those building blocks, instead of any real-number combination? (**Lattice**)

This lecture answers each of these in order. By the end, the definition of a lattice (Section 8) should feel like the natural, inevitable next step, not an isolated new idea.

## 2 Sets and Binary Operations (Quick Refresher)

A **set** is just a collection of objects (numbers, vectors, symbols — anything). A **binary operation** on a set  $S$  is a rule that takes two elements of  $S$  and combines them into a third element, e.g., ordinary addition  $+$  takes two numbers and produces their sum. We write  $a * b$  for a general binary operation.

An operation is **closed** on  $S$  if combining two elements of  $S$  always gives another element of  $S$  (never something outside  $S$ ). This sounds obvious but fails more often than you’d expect — see Example 1 below.

### Example: Closure Can Fail

Let  $S = \{1, 2, 3, 4, 5\}$  (just these five numbers) and let  $*$  be ordinary addition. Then  $2 * 4 = 6$ , but  $6 \notin S$ . So ordinary addition is *not* closed on  $S$ . This is exactly the kind of problem modular arithmetic (Section 5) is built to fix.

## 3 Groups

### Definition: Group

A set  $G$  together with a binary operation  $*$  is called a **group**, written  $(G, *)$ , if all four of the following hold:

(G1) **Closure:** for all  $a, b \in G$ ,  $a * b \in G$ .

(G2) **Associativity:** for all  $a, b, c \in G$ ,  $(a * b) * c = a * (b * c)$ .

(G3) **Identity:** there exists an element  $e \in G$  such that  $a * e = e * a = a$  for all  $a \in G$ .

(G4) **Inverses:** for every  $a \in G$ , there exists  $a^{-1} \in G$  such that  $a * a^{-1} = a^{-1} * a = e$ .

If, in addition,  $a * b = b * a$  for all  $a, b \in G$  (the order doesn’t matter), the group is called **abelian** (commutative).

### Example: Groups You Already Know

- $(\mathbb{Z}, +)$  — the integers under addition. Identity is 0; the inverse of  $a$  is  $-a$ . This is an abelian group.
- $(\mathbb{R} \setminus \{0\}, \times)$  — nonzero real numbers under multiplication. Identity is 1; the inverse of  $a$  is  $1/a$ . Also abelian. (We had to remove 0 because 0 has no multiplicative inverse.)
- $(\mathbb{Z}, \times)$  — integers under multiplication — is **not** a group: 2 has no inverse in  $\mathbb{Z}$  (we'd need  $1/2$ , which isn't an integer). This fails axiom (G4).

### Remark: Why Cryptography Cares About Groups

Many classical cryptographic schemes (Diffie–Hellman, ElGamal, elliptic-curve cryptography) are built directly on top of group structure — the security relies on certain problems (like “given  $g$  and  $g^a$ , find  $a$ ”) being hard inside a specific group. Lattice-based cryptography instead builds on richer structures (rings and modules, coming next), but the group is always the first layer underneath.

### Exercise: Try This Before Next Lecture

Is  $(\mathbb{Z}_5 \setminus \{0\}, \times \bmod 5) = (\{1, 2, 3, 4\}, \times \bmod 5)$  a group? Check all four axioms directly by building the multiplication table. (You'll need Section 6 for what “mod 5” means if it's new to you — come back to this after reading that section.)

## 3.1 Subgroups

There is one more group-related idea we need before we can properly define a lattice: a group living *inside* another group.

### Definition: Subgroup

Let  $(G, *)$  be a group. A subset  $H \subseteq G$  is a **subgroup** of  $G$  if  $H$  is itself a group under the same operation  $*$ . In practice, this is easiest to check with a simple test:  $H$  is a subgroup of  $G$  if and only if (1)  $H$  is nonempty (usually: it contains the identity  $e$ ), (2)  $H$  is closed under  $*$  (for  $a, b \in H$ ,  $a * b \in H$ ), and (3)  $H$  is closed under inverses (for  $a \in H$ ,  $a^{-1} \in H$ ). Associativity is automatic, since it already held for all of  $G$ .

### Example: Subgroups You Can Picture

- For any integer  $n$ , the set  $n\mathbb{Z} = \{\dots, -2n, -n, 0, n, 2n, \dots\}$  (all multiples of  $n$ ) is a subgroup of  $(\mathbb{Z}, +)$ : it contains 0, adding two multiples of  $n$  gives another multiple of  $n$ , and  $-(kn)$  is also a multiple of  $n$ . For example,  $2\mathbb{Z}$  is “all even integers.”
- The natural numbers  $\mathbb{N} = \{0, 1, 2, \dots\}$  are **not** a subgroup of  $(\mathbb{Z}, +)$ :  $\mathbb{N}$  is closed under addition, but it fails closure under inverses —  $1 \in \mathbb{N}$ , but  $-1 \notin \mathbb{N}$ .

### Looking Ahead: Why This Matters in Two Sections

Keep the example  $n\mathbb{Z} \subseteq \mathbb{Z}$  in mind. In Section 8, we will define a lattice as a special kind of subgroup of  $\mathbb{R}^n$  — and  $n\mathbb{Z}$  is exactly a simple, one-dimensional lattice in disguise: an evenly-spaced, discrete set of points, closed under addition and negation, sitting inside the larger group  $\mathbb{Z}$  (or  $\mathbb{R}$ ).

### 3.2 Cosets and Quotient Groups: Where $\mathbb{Z}_n$ Really Comes From

Section 6 will introduce  $\mathbb{Z}_n$  informally, as “remainders with wraparound arithmetic.” That’s the right intuition, but it’s worth seeing the fully rigorous construction now, because it is a direct, satisfying payoff of the subgroup idea — and it is the same construction (“take a group, collapse it by a subgroup”) that shows up again later when we build the ring  $R_q = \mathbb{Z}_q[x]/(x^n + 1)$  that Kyber and Dilithium actually use.

#### Definition: Coset

Let  $H$  be a subgroup of an abelian group  $(G, +)$ . For  $a \in G$ , the **coset**  $a+H$  is the set  $\{a+h : h \in H\}$  — “shift  $H$  by  $a$ .” Two elements  $a, b \in G$  turn out to produce the *same* coset,  $a+H = b+H$ , exactly when  $a-b \in H$ . This gives a clean way to sort all of  $G$  into non-overlapping “buckets,” each one a shifted copy of  $H$ .

#### Example: The Cosets of $3\mathbb{Z}$ in $\mathbb{Z}$

Take  $G = \mathbb{Z}$  and  $H = 3\mathbb{Z} = \{\dots, -3, 0, 3, 6, \dots\}$  (a subgroup, per the “Subgroups You Can Picture” example in Section 3.1). The cosets are:

- $0 + 3\mathbb{Z} = \{\dots, -3, 0, 3, 6, \dots\}$  (multiples of 3)
- $1 + 3\mathbb{Z} = \{\dots, -2, 1, 4, 7, \dots\}$  (remainder 1 when divided by 3)
- $2 + 3\mathbb{Z} = \{\dots, -1, 2, 5, 8, \dots\}$  (remainder 2 when divided by 3)

Every integer falls into exactly one of these three buckets — and which bucket it falls into is precisely its remainder mod 3. There is no fourth distinct coset:  $3 + 3\mathbb{Z} = 0 + 3\mathbb{Z}$  (check:  $3 - 0 = 3 \in 3\mathbb{Z}$ ), so the pattern repeats.

#### Definition: Quotient Group

The set of all distinct cosets of  $H$  in  $G$ , with addition defined by  $(a+H) + (b+H) := (a+b)+H$ , is itself a group — the **quotient group**, written  $G/H$  (“ $G \bmod H$ ”). Its identity element is the coset  $0 + H = H$  itself.

#### Remark: $\mathbb{Z}_n$ Is Not Just Like $\mathbb{Z}/n\mathbb{Z}$ — It Is $\mathbb{Z}/n\mathbb{Z}$

Compare the coset example above to Section 6: the three cosets of  $3\mathbb{Z}$  in  $\mathbb{Z}$  correspond *exactly* to the three elements  $\{0, 1, 2\}$  of  $\mathbb{Z}_3$ , and coset addition matches addition mod 3 exactly. This is not a coincidence or an analogy — it is a theorem:  $\mathbb{Z}_n$ , as defined informally in Section 6, is precisely the quotient group  $\mathbb{Z}/n\mathbb{Z}$ . “Wraparound arithmetic” and “sorting integers into cosets of  $n\mathbb{Z}$ ” are two descriptions of the exact same object. This is the rigorous justification for a claim we made without proof in Section 6: that  $\mathbb{Z}_n$  really is a well-defined group (in fact ring) under addition and multiplication with wraparound — it inherits this structure directly from being a quotient of  $\mathbb{Z}$ .

#### Exercise: Practice

Let  $H = 4\mathbb{Z} \subseteq \mathbb{Z}$ . List all distinct cosets of  $H$  in  $\mathbb{Z}$ , and confirm there are exactly 4 of them, matching  $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ . Which coset does  $-5$  belong to?

## 4 Rings

A ring is what you get when you ask a set to support *two* binary operations that play two structurally different roles — one behaves like addition (fully invertible), the other behaves like multiplication (weaker, no inverses required). We always *label* these two operations “+” and “×,” but — and this is worth being precise about, since it’s easy to misread — **the labels do not mean the operations have to literally be**

**school-arithmetic addition and multiplication.** They can be almost any two operations at all, as long as together they satisfy the axioms below. “+” and “×” are names for the *roles* the two operations play, not a requirement about what the operations must concretely do.

### Definition: Ring

A set  $R$  with two binary operations  $+$  and  $\times$  is a **ring**  $(R, +, \times)$  if:

(R1)  $(R, +)$  is an abelian group (with identity element usually written 0).

(R2)  $\times$  is associative:  $(a \times b) \times c = a \times (b \times c)$ .

(R3) The distributive laws hold:  $a \times (b + c) = (a \times b) + (a \times c)$  and  $(a + b) \times c = (a \times c) + (b \times c)$ .

Note: a ring does *not* need multiplicative inverses, and multiplication need not be commutative in general (though in every ring we use in this thesis, it will be — these are called **commutative rings**). If there is an element  $1 \in R$  with  $a \times 1 = 1 \times a = a$  for all  $a$ , we say  $R$  is a **ring with unity**.

### Remark: “+” and “×” Are Role Labels, Not a Restriction to Arithmetic

Compare this to groups (Section 3): a group has *one* operation, and we wrote it as a generic  $*$  precisely to emphasize that it could be anything (addition, multiplication, matrix multiplication, function composition — whatever satisfies the axioms). A ring needs *two* operations, and it turns out we always call them “+” and “×” — but this is a naming convention for the two *roles*, not a claim that the operations must be ordinary numeric addition and multiplication. Concretely:

- **$n \times n$  matrices:** “+” is ordinary matrix addition (this forms an abelian group), and “×” is matrix *multiplication* (associative, distributes over +) — but matrix multiplication is **not** commutative in general, which is fine, since a ring’s axioms above don’t require it (only *commutative* rings require it, and this one wouldn’t qualify as one).
- **Boolean rings** (used in digital logic): “+” is XOR and “×” is AND on  $\{0, 1\}$  — nothing to do with school arithmetic at all, yet every ring axiom above holds.

So whenever you meet a new ring in this course, the first question to ask is not “is + really addition and is × really multiplication?” but rather: *what are the two operations here, and do they satisfy axioms (R1)–(R3)?* The labels “+”/“×” just tell you which operation is playing which role once you’ve checked that.

### Example: Rings You Already Know

- $(\mathbb{Z}, +, \times)$  — the integers form a commutative ring with unity 1. Notice 2 still has no multiplicative inverse, and that’s fine — rings don’t require one.
- **Polynomial rings**, e.g.  $\mathbb{Z}[x]$ : the set of all polynomials with integer coefficients (like  $3x^2 - x + 7$ ), with ordinary polynomial addition and multiplication. This is a ring for exactly the same reason  $\mathbb{Z}$  is.

### Looking Ahead: The Ring That Kyber and Dilithium Actually Use

Both Kyber and Dilithium do their arithmetic inside a specific ring of the form

$$R_q = \mathbb{Z}_q[x]/(x^n + 1),$$

read as: “polynomials with coefficients taken modulo  $q$  (Section 6), where we also reduce powers of  $x$  using the rule  $x^n \equiv -1$ .” This looks intimidating right now, but it is built from exactly the pieces in this lecture: a polynomial ring (this section), with coefficients from  $\mathbb{Z}_q$  (Section 6), further “folded” by a fixed rule. We will unpack this fully in a later lecture — for now, just notice that “ring” is not an abstract curiosity; it’s the literal object the cryptography is built on.

## 4.1 Zero Divisors: A Structural Reason Some Rings Aren't Fields

We will soon see (Section 5) that  $\mathbb{Z}_n$  is a field exactly when  $n$  is prime. Section 6.1 will justify this computationally, via gcd. Here is the deeper, structural reason *why* — worth seeing now, while we're still inside general ring theory.

### Definition: Zero Divisor

In a ring  $R$ , a nonzero element  $a$  is a **zero divisor** if there exists some *nonzero*  $b \in R$  with  $a \times b = 0$ . A commutative ring with unity that has *no* zero divisors (other than needing  $a$  or  $b$  to be 0) is called an **integral domain**.

### Example: Zero Divisors in $\mathbb{Z}_6$ , Absent in $\mathbb{Z}_5$

In  $\mathbb{Z}_6$ :  $2 \times 3 = 6 \equiv 0 \pmod{6}$ , but neither 2 nor 3 is 0. So 2 and 3 are both zero divisors —  $\mathbb{Z}_6$  is *not* an integral domain. Contrast this with  $\mathbb{Z}_5$ : checking every pair of nonzero elements (1, 2, 3, 4), no product is ever  $\equiv 0 \pmod{5}$  (e.g.,  $2 \times 3 = 6 \equiv 1$ ,  $4 \times 4 = 16 \equiv 1$ , etc.) —  $\mathbb{Z}_5$  has no zero divisors.

### Remark: Why This Explains the Prime/Composite Split

**Fact:** every field is automatically an integral domain (if  $a \neq 0$  and  $ab = 0$ , multiply both sides by  $a^{-1}$  to get  $b = 0$  — so no nonzero  $b$  can pair with a nonzero  $a$  to give 0). Contrapositive: *if a ring has a zero divisor, it cannot be a field*. Now,  $\mathbb{Z}_n$  has a zero divisor exactly when  $n$  is composite: write  $n = jk$  with  $1 < j, k < n$ ; then  $j \times k = n \equiv 0 \pmod{n}$ , with neither  $j$  nor  $k$  equal to 0 in  $\mathbb{Z}_n$ . So every composite modulus produces zero divisors, which immediately rules out being a field — a purely structural explanation, with no gcd computation needed, of the same fact Section 6.1 reaches computationally.

### Exercise: Practice

Find a pair of zero divisors in  $\mathbb{Z}_{12}$  (there are several valid answers). Then explain, using the Fact above, why  $\mathbb{Z}_{12}$  cannot be a field — without computing any inverses directly.

## 5 Fields

### Definition: Field

A **field** is a set  $F$  with two binary operations  $+$  and  $\times$  such that:

(F1)  $(F, +)$  is an abelian group, over *all* of  $F$  (identity element 0).

(F2)  $(F \setminus \{0\}, \times)$  is an abelian group, over  $F$  **with 0 removed** (identity element 1).

(F3) The distributive law holds:  $a \times (b + c) = (a \times b) + (a \times c)$ .

Informally: a field is a set where you can add, subtract, multiply, and divide (by anything except 0) and always stay inside the set. In short — **a field is two abelian groups sharing one set, glued together by distributivity**.

### Remark: Why 0 Has to Be Excluded, and It's Not Optional

In *any* structure with a distributive law,  $0 \times x = 0$  for every  $x$  (this follows from distributivity itself, not from an extra rule:  $0 \times x = (0 + 0) \times x = 0 \times x + 0 \times x$ , so subtracting  $0 \times x$  from both sides gives  $0 = 0 \times x$ ). So 0 could only have a multiplicative inverse if  $0 \times x = 1$  for some  $x$  — but the left side is always 0, never 1 (as long as  $1 \neq 0$ , which we always assume). So 0 is permanently excluded from having an inverse, in every field, without exception; it isn't a special case someone chose, it's forced by the arithmetic.

### Definition: Subtraction and Division Are *Derived*, Not Extra

Neither subtraction nor division is a new, independent operation requiring its own axiom — both are just notation for “combine with an inverse,” one level for each operation:

$$a - b := a + (-b), \quad \frac{a}{b} := a \times b^{-1} \quad (b \neq 0),$$

where  $-b$  is guaranteed to exist by  $(F, +)$  being a group (F1), and  $b^{-1}$  is guaranteed to exist, for every *nonzero*  $b$ , by  $(F \setminus \{0\}, \times)$  being a group (F2). So “a field supports division” is not an extra requirement on top of the definition — it falls straight out of (F2) once every nonzero element is already promised an inverse. This is exactly why (F2) excludes 0: since 0 can never have an inverse (see the remark above), “dividing by 0” can never be defined this way, no matter what field you're in.

### Example: Division in $\mathbb{Z}_5$ , Worked Out

What is  $4 \div 3$  in  $\mathbb{Z}_5$ ? First find  $3^{-1} \bmod 5$ :  $3 \times 2 = 6 \equiv 1 \pmod{5}$ , so  $3^{-1} = 2$ . Then  $4 \div 3 := 4 \times 3^{-1} = 4 \times 2 = 8 \equiv 3 \pmod{5}$ . Check:  $3 \times 3 = 9 \equiv 4 \pmod{5}$  — correct, exactly as “ $4 \div 3 = 3$ ” should verify.

### Remark: The One-Line Takeaway

**Subtraction is the additive inverse; division is the multiplicative inverse.** Same trick, applied once per operation —  $a - b$  means “add  $a$  and  $b$ 's additive inverse,”  $a/b$  means “multiply  $a$  by  $b$ 's multiplicative inverse.” The one asymmetry between them traces directly back to (F1) vs. (F2): *every* element of  $F$  has an additive inverse, since  $(F, +)$  is a group over *all* of  $F$  — so subtraction always works. But *only nonzero* elements have a multiplicative inverse, since  $(F \setminus \{0\}, \times)$  is a group only after removing 0 — so division by  $b$  only ever works when  $b \neq 0$ .

**Every Field Is a Ring** Every field  $F$  (Definition above) is automatically a commutative ring with unity — in fact, an **integral domain** (Section 4.1). *Why*: (F1) is exactly ring axiom (R1). Associativity and distributivity of  $\times$  (ring axioms (R2)–(R3)) are inherited directly from  $\times$  being a group operation on  $F \setminus \{0\}$  together with (F3). So a field satisfies every ring axiom, plus two extra guarantees stacked on top:  $\times$  is commutative, and every *nonzero* element is invertible. This is why some textbooks instead *define* a field as “a commutative ring with unity in which every nonzero element has a multiplicative inverse” — that phrasing and the definition above pick out exactly the same objects; it's just built in the reverse order (starting from “ring” and adding a condition, instead of starting from two groups and gluing them together). Finally, a field is automatically an integral domain too: if  $ab = 0$  with  $a \neq 0$ , multiplying both sides by  $a^{-1}$  forces  $b = a^{-1}(ab) = a^{-1} \cdot 0 = 0$ , so there can be no zero divisors.



### Example: Fields and Non-Fields

- $\mathbb{Q}$  (rationals),  $\mathbb{R}$  (reals),  $\mathbb{C}$  (complex numbers) are all fields — you can divide by any nonzero element and stay inside the set.
- $\mathbb{Z}$  is **not** a field:  $2 \in \mathbb{Z}$  has no multiplicative inverse in  $\mathbb{Z}$  (again,  $1/2 \notin \mathbb{Z}$ ).  $\mathbb{Z}$  is a ring, but not a field.
- $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$  under addition and multiplication mod 5 *is* a field — every nonzero element has an inverse (you checked this, or will check this, in the Section 3 subgroup/group exercise). Fields built this way, from  $\mathbb{Z}_p$  with  $p$  prime, are called **finite fields** or **Galois fields**, written  $\mathbb{F}_p$  or  $GF(p)$ .

### Remark: The One-Sentence Summary So Far

Group  $\subset$  Ring  $\subset$  Field, in the sense that a field is a ring with extra structure (division), and a ring is built from a group with extra structure (a second operation). Each step adds a capability: groups let you undo one operation (inverses); rings add a second, distributive operation; fields guarantee *both* operations are fully invertible (except division by 0).

## 6 The Modulus: Arithmetic That Wraps Around

### Definition: Congruence Modulo $n$

For a fixed positive integer  $n$  (the **modulus**), two integers  $a$  and  $b$  are **congruent modulo  $n$** , written

$$a \equiv b \pmod{n},$$

if  $n$  divides  $(a - b)$  evenly — equivalently,  $a$  and  $b$  leave the same remainder when divided by  $n$ . The set of possible remainders,  $\{0, 1, 2, \dots, n - 1\}$ , together with addition and multiplication performed “with wraparound,” is written  $\mathbb{Z}_n$  (or  $\mathbb{Z}/n\mathbb{Z}$ ).

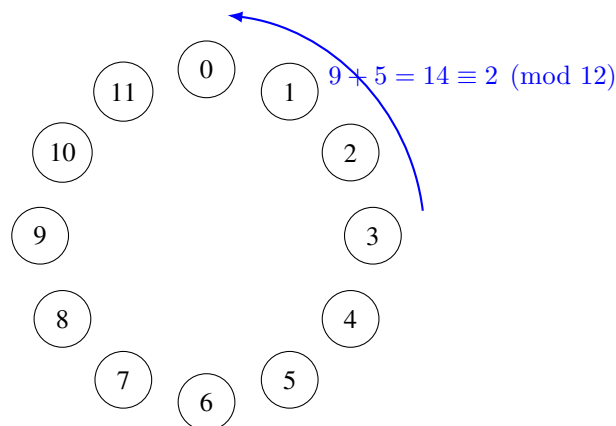


Figure 1: The clock analogy for modular arithmetic:  $\mathbb{Z}_{12}$  works exactly like a 12-hour clock face. 9 o'clock plus 5 hours wraps around to 2 o'clock, i.e.,  $9 + 5 = 14 \equiv 2 \pmod{12}$ .

### Example: Computing Modulo

$17 \bmod 5$ : divide 17 by 5 to get 3 remainder 2, so  $17 \equiv 2 \pmod{5}$ . Similarly,  $23 \equiv 3 \pmod{10}$ , and  $-1 \equiv n - 1 \pmod{n}$  (e.g.,  $-1 \equiv 4 \pmod{5}$ ) — negative numbers wrap around too.



### Remark: When Is $\mathbb{Z}_n$ a Ring? When Is It a Field?

$\mathbb{Z}_n$  is *always* a commutative ring with unity, for any  $n \geq 2$ . But it is a *field* only when  $n = p$  is **prime**. If  $n$  is composite (e.g.,  $n = 6$ ), some nonzero elements fail to have multiplicative inverses — e.g., in  $\mathbb{Z}_6$ , there is no  $x$  with  $2x \equiv 1 \pmod{6}$ , since 2 shares a common factor with 6. This is exactly why cryptographic schemes that need a field (rather than just a ring) are built with a prime modulus  $q$ . (Section 4.1 gave the deeper structural reason: composite  $n$  always produces zero divisors, and a ring with zero divisors can never be a field.)

## 6.1 Finding Inverses Without Guessing: The Extended Euclidean Algorithm

So far, the only way we’ve found a multiplicative inverse mod  $n$  is trial and error (Exercise: “find  $x$  with  $3x \equiv 1 \pmod{7}$ ” — you could just try  $x = 1, 2, 3, \dots$  until one works). That’s fine for small numbers, but real cryptographic moduli are enormous (Kyber’s modulus is  $q = 3329$ ; RSA moduli have hundreds of digits) — trial and error is completely impractical. We need an actual algorithm.

### Definition: Greatest Common Divisor (GCD)

For integers  $a, b$  (not both zero),  $\gcd(a, b)$  is the largest positive integer that divides both  $a$  and  $b$ . Two integers are **coprime** (or relatively prime) if  $\gcd(a, b) = 1$ .

### Remark: Fact: When Does an Inverse Exist?

An element  $a \in \mathbb{Z}_n$  has a multiplicative inverse if and only if  $\gcd(a, n) = 1$ . This is why every nonzero element of  $\mathbb{Z}_p$  (for prime  $p$ ) has an inverse: any  $a$  with  $1 \leq a \leq p - 1$  automatically satisfies  $\gcd(a, p) = 1$ , since  $p$ ’s only divisors are 1 and  $p$  itself.

The classical **Euclidean Algorithm** computes  $\gcd(a, b)$  by repeated division with remainder:  $\gcd(a, b) = \gcd(b, a \bmod b)$ , stopping when the remainder hits 0. The **Extended Euclidean Algorithm** does this same process but also tracks, at *every single step*, how the current remainder can be written as a combination  $a \cdot x + b \cdot y$  (integers  $x, y$ ) — and when the algorithm finishes with  $\gcd(a, b) = 1$ , the row where the remainder equals 1 hands you the inverse directly. The cleanest way to track this — much easier to follow than plain back-substitution — is a running **table** with one row per step.

### Definition: The Extended Euclidean Algorithm, as a Table

To find  $\gcd(a, b)$  together with integers  $x, y$  satisfying  $a \cdot x + b \cdot y = \gcd(a, b)$ , build a table with columns  $q$  (quotient),  $r$  (remainder),  $x, y$ , starting with two fixed initial rows and then one new row per division step:

| step     | $q$                                | $r$                      | $x$                      | $y$                      |
|----------|------------------------------------|--------------------------|--------------------------|--------------------------|
| −1       | —                                  | $a$                      | 1                        | 0                        |
| 0        | —                                  | $b$                      | 0                        | 1                        |
| 1        | $q_1 = \lfloor r_{-1}/r_0 \rfloor$ | $r_1 = r_{-1} - q_1 r_0$ | $x_1 = x_{-1} - q_1 x_0$ | $y_1 = y_{-1} - q_1 y_0$ |
| 2        | $q_2 = \lfloor r_0/r_1 \rfloor$    | $r_2 = r_0 - q_2 r_1$    | $x_2 = x_0 - q_2 x_1$    | $y_2 = y_0 - q_2 y_1$    |
| $\vdots$ | $\vdots$                           | $\vdots$                 | $\vdots$                 | $\vdots$                 |

Every single row satisfies the **invariant**  $r_i = a \cdot x_i + b \cdot y_i$  by construction — this is the whole mechanism, and it’s worth checking explicitly on at least one row the first time you use this table, to convince yourself it isn’t magic. Stop once a row produces  $r_i = 0$ ; the *previous* row’s  $r$  is  $\gcd(a, b)$ , and that row’s  $x, y$  are exactly the combination you want.

**Example: Table Method, Short Version:  $\gcd(7, 3)$  and  $3^{-1} \bmod 7$**

| step | $q$                       | $r$            | $x$             | $y$             | check: $7x + 3y$              |
|------|---------------------------|----------------|-----------------|-----------------|-------------------------------|
| -1   | —                         | 7              | 1               | 0               | $7(1) + 3(0) = 7 \checkmark$  |
| 0    | —                         | 3              | 0               | 1               | $7(0) + 3(1) = 3 \checkmark$  |
| 1    | $\lfloor 7/3 \rfloor = 2$ | $7 - 2(3) = 1$ | $1 - 2(0) = 1$  | $0 - 2(1) = -2$ | $7(1) + 3(-2) = 1 \checkmark$ |
| 2    | $\lfloor 3/1 \rfloor = 3$ | $3 - 3(1) = 0$ | $0 - 3(1) = -3$ | $1 - 3(-2) = 7$ | (stop: $r = 0$ )              |

The row *before*  $r$  hit 0 is step 1:  $r_1 = 1 = \gcd(7, 3)$ , with  $x_1 = 1, y_1 = -2$ . So  $7(1) + 3(-2) = 1$ , i.e.,  $3 \times (-2) \equiv 1 \pmod{7}$ , giving  $3^{-1} \equiv -2 \equiv 5 \pmod{7}$  — matching what trial and error would also find, but now via a method with a clear, mechanical stopping point.

**Example: Table Method, Longer Version:  $15^{-1} \bmod 26$**

Here the table earns its keep — more steps than the tiny example above, but the process is identical, row by row.

| step | $q$                         | $r$               | $x$              | $y$               |
|------|-----------------------------|-------------------|------------------|-------------------|
| -1   | —                           | 26                | 1                | 0                 |
| 0    | —                           | 15                | 0                | 1                 |
| 1    | $\lfloor 26/15 \rfloor = 1$ | $26 - 1(15) = 11$ | $1 - 1(0) = 1$   | $0 - 1(1) = -1$   |
| 2    | $\lfloor 15/11 \rfloor = 1$ | $15 - 1(11) = 4$  | $0 - 1(1) = -1$  | $1 - 1(-1) = 2$   |
| 3    | $\lfloor 11/4 \rfloor = 2$  | $11 - 2(4) = 3$   | $1 - 2(-1) = 3$  | $-1 - 2(2) = -5$  |
| 4    | $\lfloor 4/3 \rfloor = 1$   | $4 - 1(3) = 1$    | $-1 - 1(3) = -4$ | $2 - 1(-5) = 7$   |
| 5    | $\lfloor 3/1 \rfloor = 3$   | $3 - 3(1) = 0$    | $3 - 3(-4) = 15$ | $-5 - 3(7) = -26$ |

The row before  $r$  hit 0 is step 4:  $r_4 = 1 = \gcd(26, 15)$ , with  $x_4 = -4, y_4 = 7$ . **Check the invariant directly:**  $26(-4) + 15(7) = -104 + 105 = 1 \checkmark$ . Reading this modulo 26:  $15 \times 7 \equiv 1 \pmod{26}$ , so  $15^{-1} \equiv 7 \pmod{26}$ . **Final check:**  $15 \times 7 = 105 = 4 \times 26 + 1 \equiv 1 \pmod{26}$ . Correct.

**Exercise: Practice**

Use the table method above (not back-substitution, and not trial and error) to find  $5^{-1} \bmod 11$ . Build the full table — steps  $-1, 0, 1, 2, \dots$  — and stop at the correct row. Then check your answer by direct multiplication.

**Looking Ahead: Why Cryptography Loves the Modulus**

Modular arithmetic gives us two things cryptography needs badly: (1) a **finite** number system (so keys and computations have a fixed, bounded size, no matter how much you compute), and (2) built-in **one-wayness** in places — e.g., given only  $a \bmod n$ , you cannot tell which of infinitely many integers  $a$  actually was. The Learning-With-Errors (LWE) problem from the proposal document hides its secret inside equations that live in exactly this kind of finite, modular number system (there, the modulus is usually called  $q$ ).

**Exercise: Practice**

(a) Compute  $23 \times 15 \bmod 7$ . (b) List all elements of  $\mathbb{Z}_8$  that *do not* have a multiplicative inverse, and explain why (hint: compare each element to 8). (c) Confirm  $\mathbb{Z}_7$  is a field by checking that 3 has a multiplicative inverse mod 7 (find  $x$  such that  $3x \equiv 1 \pmod{7}$ ).

## 7 Vector Spaces, Basis, and Dimension

### Definition: Vector Space

A **vector space** over a field  $F$  is a set  $V$  (whose elements are called **vectors**) equipped with vector addition and scalar multiplication (multiplying a vector by an element of  $F$ ), satisfying the “obvious” rules you’d expect: addition makes  $(V, +)$  an abelian group, and scalar multiplication distributes sensibly over both vector addition and field addition. The most important example for us:  $\mathbb{R}^n$ , the set of all  $n$ -tuples of real numbers, is a vector space over  $\mathbb{R}$ .

### Definition: Linear Combination, Span, Linear Independence

Given vectors  $v_1, \dots, v_k \in V$  and scalars  $c_1, \dots, c_k \in F$ , the vector  $c_1v_1 + c_2v_2 + \dots + c_kv_k$  is called a **linear combination** of  $v_1, \dots, v_k$ . The set of *all* linear combinations of a set of vectors is called their **span**. The vectors  $v_1, \dots, v_k$  are **linearly independent** if the only way to write  $0 = c_1v_1 + \dots + c_kv_k$  is with every  $c_i = 0$  — informally, no vector in the set is “redundant” (expressible using the others).

### Definition: Basis and Dimension

A set of vectors  $B = \{b_1, \dots, b_k\}$  is a **basis** of a vector space  $V$  if (1)  $B$  is linearly independent, and (2)  $B$  **spans**  $V$  (every vector in  $V$  can be written as a linear combination of  $B$ ). Every basis of a given vector space has the same number of elements  $k$ , called the **dimension** of  $V$ .

### Example: Basis of $\mathbb{R}^2$

The vectors  $e_1 = (1, 0)$  and  $e_2 = (0, 1)$  form the “standard” basis of  $\mathbb{R}^2$ : any vector  $(a, b)$  is exactly  $a \cdot e_1 + b \cdot e_2$ . But this is far from the *only* basis —  $\{(1, 1), (1, -1)\}$  is also a valid basis of  $\mathbb{R}^2$  (check: they’re linearly independent, and any  $(a, b)$  can be written as a combination of them). A vector space almost always has infinitely many possible bases; some are more convenient to work with than others. **Keep this fact in mind — it becomes very important in Section 8.**

## 7.1 Inner Products, Norms, and Orthogonality

To make “good basis” (short, nearly-perpendicular vectors) versus “bad basis” (long, nearly-parallel vectors) precise rather than just intuitive, we need a way to measure length and angle.

### Definition: Inner Product and Norm

For  $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n) \in \mathbb{R}^n$ , the (standard) **inner product** is  $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$ . The **norm** (length) of  $x$  is  $\|x\| = \sqrt{\langle x, x \rangle}$ . Two vectors are **orthogonal** if  $\langle x, y \rangle = 0$  (the angle between them is  $90^\circ$ ). A basis  $\{b_1, \dots, b_k\}$  is an **orthogonal basis** if every pair  $b_i, b_j$  ( $i \neq j$ ) is orthogonal.

### Example: Computing Norms and Checking Orthogonality

For  $x = (3, 4)$ :  $\|x\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$ . For  $x = (1, 1)$  and  $y = (1, -1)$ :  $\langle x, y \rangle = (1)(1) + (1)(-1) = 0$ , so these two vectors (the alternate basis of  $\mathbb{R}^2$  from the previous example) are orthogonal — a nicer basis, geometrically, than it might have looked at first glance.

### Looking Ahead: Quantifying “Good” vs. “Bad” Bases

With norms in hand, we can finally give a real, measurable meaning to the “good basis / bad basis” idea flagged repeatedly in this lecture: a good basis has vectors with small norm  $\|b_i\|$  that are close to orthogonal; a bad basis for the *same* lattice can have arbitrarily large norms and near-parallel vectors, despite generating an identical set of points. One common numerical measure of “how good” a basis is is its **orthogonality defect**,  $\frac{\prod_i \|b_i\|}{\det(L)}$  (see Section 8.4 below for  $\det(L)$ ) — it equals 1 for a perfectly orthogonal basis and grows the more skewed the basis is.

## 7.2 Gram–Schmidt Orthogonalization

Given *any* basis, can we systematically produce an orthogonal one (spanning the same space, though generally *not* the same lattice — more on that distinction below)? Yes:

### Definition: Gram–Schmidt Process

Given linearly independent  $b_1, \dots, b_k$ , define  $b_1^* = b_1$ , and for  $i = 2, \dots, k$ :

$$b_i^* = b_i - \sum_{j=1}^{i-1} \mu_{i,j} b_j^*, \quad \text{where } \mu_{i,j} = \frac{\langle b_i, b_j^* \rangle}{\langle b_j^*, b_j^* \rangle}.$$

Each  $b_i^*$  is  $b_i$  with the “shadow” it casts on all previous  $b_j^*$  subtracted off, leaving only the genuinely new, orthogonal direction. The result  $\{b_1^*, \dots, b_k^*\}$  is an orthogonal basis of the same vector space (though, importantly, its integer combinations generally do *not* produce the same lattice as the original  $b_i$ ’s — Gram–Schmidt vectors are a vector-space tool, not automatically a lattice basis).

### Example: Gram–Schmidt on a $2 \times 2$ Example

Let  $b_1 = (2, 0)$ ,  $b_2 = (1, 2)$ . Then  $b_1^* = b_1 = (2, 0)$ . Compute  $\mu_{2,1} = \frac{\langle b_2, b_1^* \rangle}{\langle b_1^*, b_1^* \rangle} = \frac{(1)(2) + (2)(0)}{4} = \frac{2}{4} = 0.5$ . So  $b_2^* = b_2 - 0.5 b_1^* = (1, 2) - (1, 0) = (0, 2)$ . Check:  $\langle b_1^*, b_2^* \rangle = (2)(0) + (0)(2) = 0$  — orthogonal, as required.

### Looking Ahead: Where Gram–Schmidt Leads

This process reappears, in a modified “integer-preserving” form, inside the **LLL algorithm** — the classic algorithm for turning a bad lattice basis into a reasonably good one, and a core tool used both to *attack* weak lattice-based schemes and to reason about the security of well-parametrized ones. We’ll meet it properly in a later lecture; for now, just remember that Gram–Schmidt is the geometric heart of it.

### Exercise: Practice

Run Gram–Schmidt on  $b_1 = (1, 1)$ ,  $b_2 = (0, 2)$ : compute  $b_1^*$ ,  $\mu_{2,1}$ , and  $b_2^*$ , and verify  $\langle b_1^*, b_2^* \rangle = 0$ .

## 8 Lattices, Properly Defined

Everything so far has allowed *any* scalar from the field  $F$  (e.g., any real number) when forming linear combinations. Informally, a lattice is what happens when we impose one dramatic restriction: **only integer combinations are allowed**, which turns a solid, continuous vector space into a discrete grid of isolated points. That informal picture (Figure 3 below) is correct and worth keeping in your head. But the definition we will actually use — the one used in serious treatments of the subject, such as

Chris Peikert's graduate lecture notes *Lattices in Cryptography*<sup>1</sup> — starts from the **subgroup** idea from Section 3.1, not from a basis. We will show the two views agree.

## 8.1 Discrete Sets

### Definition: Discrete Subset

A subset  $S \subseteq \mathbb{R}^n$  is **discrete** if every point of  $S$  is *isolated*: for every  $x \in S$ , there exists some  $\varepsilon > 0$  such that the open ball  $B(x, \varepsilon)$  (all points within distance  $\varepsilon$  of  $x$ ) contains *no* point of  $S$  other than  $x$  itself. Informally: you can always draw a small enough circle around any point of  $S$  that catches nothing else from  $S$ .

### Remark: A Shortcut for Subgroups

Checking every single point of a set for isolation sounds like a lot of work. But if  $H$  is a *subgroup* of  $(\mathbb{R}^n, +)$  (Section 3.1), there's a shortcut:  $H$  is discrete if and only if the origin  $\vec{0}$  alone is isolated, i.e., there is some  $\varepsilon > 0$  with  $B(\vec{0}, \varepsilon) \cap H = \{\vec{0}\}$ . *Why this works*: suppose  $\vec{0}$  is isolated with radius  $\varepsilon$ , and suppose some other point  $y \in H$  were within  $\varepsilon$  of some  $x \in H$ . Since  $H$  is a subgroup,  $y - x \in H$  as well, and  $y - x$  is within  $\varepsilon$  of  $\vec{0}$  — so by our assumption,  $y - x = \vec{0}$ , i.e.,  $y = x$ . So checking one point (the origin) is enough for the whole subgroup, thanks to closure under subtraction.

## 8.2 The Definition

### Definition: Lattice

A **lattice** is a discrete additive subgroup  $L$  of  $\mathbb{R}^n$ . That is,  $L \subseteq \mathbb{R}^n$  is a lattice if:

- (L1)  $L$  is a subgroup of  $(\mathbb{R}^n, +)$  (Section 3.1): it contains  $\vec{0}$ , and is closed under addition and negation.
- (L2)  $L$  is discrete: by the shortcut above, there exists  $\varepsilon > 0$  such that  $B(\vec{0}, \varepsilon) \cap L = \{\vec{0}\}$ .

### Example: $\mathbb{Q}$ Is a Subgroup, But *Not* a Lattice

Consider  $\mathbb{Q} \subseteq \mathbb{R}$  (the rational numbers) under addition. This *is* a subgroup: it contains 0, the sum of two rationals is rational, and  $-a$  is rational whenever  $a$  is. So condition (L1) holds. But condition (L2) fails badly: the rationals are **dense** in  $\mathbb{R}$ , meaning every open ball around 0, no matter how tiny, contains infinitely many other rational numbers. There is no  $\varepsilon > 0$  that isolates 0. So  $\mathbb{Q}$  is a perfectly good subgroup that is **not** a lattice — this is exactly why condition (L2) has to be part of the definition; subgroup alone is not a strong enough requirement.

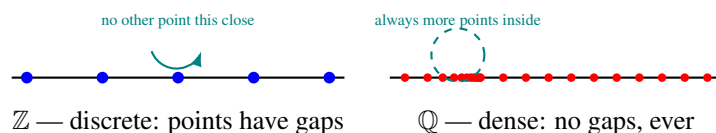


Figure 2: Discreteness, visualized. Left: in  $\mathbb{Z}$ , every point has a gap of size 1 around it with nothing else nearby — discrete. Right: in  $\mathbb{Q}$ , no matter how small a circle you draw around a point, there are always infinitely many more rational points crammed inside it — dense, and therefore *not* discrete.

<sup>1</sup>C. Peikert, *Lattices in Cryptography*, lecture notes, <https://github.com/cpeikert/LatticesInCryptography>.

### Example: $\mathbb{Z}$ and $\mathbb{Z}^n$ Are Lattices

$\mathbb{Z} \subseteq \mathbb{R}$  is a subgroup (trivially — and recall from Section 3.1 that  $n\mathbb{Z}$  is a subgroup of  $\mathbb{Z}$  for any  $n$ ; here we’re just looking at  $\mathbb{Z}$  itself sitting inside  $\mathbb{R}$ ), and it is discrete: taking  $\varepsilon = 0.5$ ,  $B(0, 0.5) \cap \mathbb{Z} = \{0\}$ , since the nearest other integers ( $\pm 1$ ) are a full unit away. So  $\mathbb{Z}$  is a lattice in  $\mathbb{R}$  — the simplest possible example. The same argument, coordinate by coordinate, shows  $\mathbb{Z}^n \subseteq \mathbb{R}^n$  (all integer-coordinate points) is a lattice in  $\mathbb{R}^n$ ; it is often called **the integer lattice**.

### 8.3 Recovering the Basis Picture

The subgroup definition is more general and more standard, but the “grid generated by a basis” picture you may already have in your head is not wrong — it is a theorem, not a definition, and it’s worth seeing why.

**Basis Representation of Lattices** Let  $L \subseteq \mathbb{R}^n$ . Then  $L$  is a lattice (in the sense of the subgroup definition above, i.e., a discrete additive subgroup) **if and only if** there exist linearly independent vectors  $b_1, \dots, b_k \in \mathbb{R}^n$  (with  $k \leq n$ ) such that

$$L = \mathcal{L}(b_1, \dots, b_k) = \left\{ \sum_{i=1}^k a_i b_i \mid a_i \in \mathbb{Z} \right\}.$$

The vectors  $b_1, \dots, b_k$  are called a **basis** of  $L$ ; the number  $k$  is the **rank** of  $L$ . If  $k = n$  (the basis spans all of  $\mathbb{R}^n$ ),  $L$  is called **full-rank** — this is the common case in cryptography.

#### Remark: Proof Idea (Sketch Only)

**Easy direction** ( $\Leftarrow$ ): given linearly independent  $b_1, \dots, b_k$ , the set of integer combinations is clearly a subgroup (adding two integer combinations, or negating one, gives another integer combination). Discreteness holds because the  $b_i$  point in genuinely independent directions: you cannot make  $\sum a_i b_i$  arbitrarily close to  $\vec{0}$  without every  $a_i$  being exactly 0 — linear independence rules out any “near-cancellation.” (Making this fully rigorous uses the fact that the linear map  $(a_1, \dots, a_k) \mapsto \sum a_i b_i$  has a bounded inverse on its image.)

**Harder direction** ( $\Rightarrow$ ): given only that  $L$  is a discrete subgroup, we must *construct* a basis. The idea is greedy and inductive: pick  $b_1$  to be a shortest nonzero vector in  $L$  (this exists precisely *because*  $L$  is discrete — in a non-discrete set like  $\mathbb{Q}$ , there is no shortest nonzero element, since you can always find a smaller one). Then look at the part of  $L$  not lying on the line through  $b_1$ , and pick  $b_2$  to be a shortest vector there, and so on. A careful argument (using discreteness at each step, and projecting onto the space orthogonal to the vectors already chosen) shows this process terminates with a valid basis after  $k \leq n$  steps. The full details are carried out rigorously in Peikert’s notes; we take the result as given here.

#### Looking Ahead: A Name for the “Shortest Vector”

The shortest nonzero vector used in the proof sketch above has a standard name:  $\lambda_1(L)$ , the **minimum distance** of the lattice. Computing  $\lambda_1(L)$  — or even just estimating it — for a lattice given only a (possibly bad) basis is called the **Shortest Vector Problem (SVP)**, and it is believed to be extremely hard in high dimensions. This hardness is one of the two or three core hard problems that lattice-based cryptography’s security ultimately rests on. We’ll return to this once we reach the LWE problem.

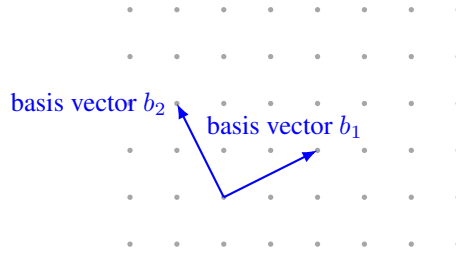


Figure 3: A 2-dimensional, full-rank lattice generated by a basis  $\{b_1, b_2\}$ : every grid dot is reachable by some *integer* combination  $a_1b_1 + a_2b_2$  (with  $a_1, a_2 \in \mathbb{Z}$ ), starting from any chosen origin dot. This is the same object as the subgroup definition above — just viewed through its basis instead of as an abstract subgroup.

#### Remark: Same Lattice, Different Basis

A single lattice has *many* different valid bases (just as a vector space does — recall the “Basis of  $\mathbb{R}^2$ ” example in Section 7). Two bases generate the exact same set of points if and only if one can be converted to the other using an integer matrix with determinant  $\pm 1$  (a **unimodular** transformation) — we won’t need the details of this fact yet, but the consequence is critical: some bases consist of short, nearly-perpendicular vectors (a “good” basis, easy to compute with), while other, equally valid bases for the *same* lattice consist of long, nearly-parallel vectors (a “bad” basis, hard to compute with). **This gap between good and bad bases for the same lattice is the single most important idea in lattice-based cryptography:** the secret key is typically a good (short) basis, while the public key describes the same lattice using a bad (long, scrambled-looking) basis. Recovering the good basis from the bad one is believed to be extremely hard once the dimension  $n$  is large enough.

#### Example: A Tiny Lattice by Hand

Let  $b_1 = (2, 0)$  and  $b_2 = (0, 3)$  in  $\mathbb{R}^2$ . The lattice  $\mathcal{L}(b_1, b_2)$  consists of every point  $(2a_1, 3a_2)$  for integers  $a_1, a_2$  — so it includes  $(0, 0)$ ,  $(2, 0)$ ,  $(-2, 0)$ ,  $(0, 3)$ ,  $(2, 3)$ ,  $(4, -3)$ , and so on, but *not*  $(1, 1)$  or  $(1, 0)$ , since those cannot be written as  $(2a_1, 3a_2)$  for integer  $a_1, a_2$ . As a sanity check against the subgroup definition directly: is this set a subgroup? Yes — closed under addition and negation, contains  $(0, 0)$ . Is it discrete? Yes — the nearest point to the origin is at distance 2, so any  $\varepsilon < 2$  isolates  $\vec{0}$ .

### 8.4 The Determinant (Volume) of a Lattice

One more piece of structure will matter a great deal in later lectures: every lattice has a well-defined “volume,” independent of which basis you use to describe it.

#### Definition: Fundamental Domain and Determinant

Given a basis  $b_1, \dots, b_n$  of a full-rank lattice  $L \subseteq \mathbb{R}^n$ , the **fundamental parallelepiped** is

$$\mathcal{P}(B) = \left\{ \sum_{i=1}^n t_i b_i \mid 0 \leq t_i < 1 \right\},$$

i.e., the region “tiled out” by taking all *fractional* (not just integer) combinations between 0 and 1. Copies of  $\mathcal{P}(B)$ , shifted by every lattice point, tile all of  $\mathbb{R}^n$  with no gaps or overlaps. The **determinant** of the lattice,  $\det(L)$ , is the volume of  $\mathcal{P}(B)$ , computed as  $\det(L) = |\det(B)|$ , where  $B$  is the matrix whose rows (or columns) are  $b_1, \dots, b_n$ .



### Remark: Why $\det(L)$ Doesn't Depend on Which Basis You Pick

Recall from the “Same Lattice, Different Basis” remark above: any two bases of the same lattice are related by a unimodular matrix  $U$  (integer entries,  $\det(U) = \pm 1$ ), i.e.,  $B' = UB$ . Then  $\det(B') = \det(U) \det(B) = \pm \det(B)$ , so  $|\det(B')| = |\det(B)|$ . This confirms  $\det(L)$  is a genuine property of the lattice itself, not an artifact of which basis happened to be used to compute it — exactly the kind of basis-independence you should expect from something called “the volume of the lattice.”

### Example: Computing $\det(L)$ for the Tiny Lattice

For  $b_1 = (2, 0)$ ,  $b_2 = (0, 3)$  from the example above,  $B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ , so  $\det(L) = |\det(B)| = |2 \times 3 - 0 \times 0| = 6$ . Geometrically: the fundamental parallelepiped is just the  $2 \times 3$  rectangle  $[0, 2) \times [0, 3)$ , and copies of this rectangle exactly tile the plane, anchored at every lattice point — matching the picture in Figure 3's style, just for this specific (axis-aligned) lattice.

### Looking Ahead: Minkowski's Theorem: Connecting Volume Back to $\lambda_1(L)$

Here is a genuinely deep fact we will only preview, not prove: **Minkowski's First Theorem** says that any symmetric convex region of volume greater than  $2^n \det(L)$  must contain a nonzero lattice point. Applying this to a ball gives a bound relating the two quantities we've now defined:  $\lambda_1(L) \leq \sqrt{n} \cdot \det(L)^{1/n}$  (up to constants). In plain terms: a lattice cannot “hide” an arbitrarily long shortest vector if its volume is small — volume and minimum distance are locked together by the geometry, no matter which basis you're handed. This is part of *why* SVP is believed hard only in the “regime” cryptographers carefully choose (high dimension  $n$ , appropriately scaled  $\det(L)$ ) rather than for arbitrary lattices — a preview of parameter selection, which we'll return to once we discuss how Kyber and Dilithium choose their concrete dimensions and moduli.

### Exercise: Practice

For the lattice generated by  $b_1 = (1, 2)$ ,  $b_2 = (3, 1)$  (from the earlier homework exercise): compute  $\det(L)$  directly from the basis matrix. Then pick any *other* valid basis for the same lattice (e.g.,  $b'_1 = b_1$ ,  $b'_2 = b_2 - b_1$ ) and confirm you get the same  $\det(L)$ .

### Looking Ahead: Where This Leads Next Lecture

With Groups  $\rightarrow$  Subgroups  $\rightarrow$  Rings  $\rightarrow$  Fields  $\rightarrow$  Modular arithmetic  $\rightarrow$  Vector spaces  $\rightarrow$  Basis  $\rightarrow$  Lattice now in place, the next lecture can finally state the **Learning With Errors (LWE)** problem precisely: a secret vector, hidden inside noisy linear equations over  $\mathbb{Z}_q$  (a ring, Section 6), which is exactly the kind of “bad basis” hardness described above, dressed up in a specific, standardized cryptographic form. We will also unpack the ring  $R_q = \mathbb{Z}_q[x]/(x^n + 1)$  flagged in Section 4, since **module lattices** (lattices whose coordinates come from a ring like  $R_q$  instead of plain integers) are exactly what Kyber and Dilithium use in practice.

## 9 Summary Table

| Structure              | Requires                                     | Example                                             |
|------------------------|----------------------------------------------|-----------------------------------------------------|
| Group                  | 1 operation, inverses exist                  | $(\mathbb{Z}, +)$                                   |
| Subgroup               | A subset that is itself a group              | $2\mathbb{Z} \subseteq \mathbb{Z}$                  |
| Coset / Quotient group | $G$ sorted into shifted copies of $H$        | $\mathbb{Z}/3\mathbb{Z} = \{0, 1, 2\}$              |
| Ring                   | 2 operations, no division needed             | $(\mathbb{Z}, +, \times), \mathbb{Z}[x]$            |
| Integral domain        | Ring, no zero divisors                       | $\mathbb{Z}, \mathbb{Z}_5$ (not $\mathbb{Z}_6$ )    |
| Field                  | Ring + division by nonzero elements          | $\mathbb{Q}, \mathbb{R}, \mathbb{Z}_p$ ( $p$ prime) |
| Modulus                | Wraparound arithmetic                        | $\mathbb{Z}_n = \{0, \dots, n-1\}$                  |
| Vector space           | Field + vectors + scalar mult.               | $\mathbb{R}^n$                                      |
| Basis                  | Independent, spanning vector set             | $\{(1, 0), (0, 1)\}$ in $\mathbb{R}^2$              |
| Orthogonal basis       | Basis, all pairs perpendicular               | Gram–Schmidt output                                 |
| Lattice                | Discrete additive subgroup of $\mathbb{R}^n$ | $\mathbb{Z}^n$ ; grid of points                     |
| $\det(L)$              | Volume of the fundamental domain             | $ \det(B) $ , any basis $B$                         |

## 10 Full List of Exercises (Assign as Homework Before Lecture 2)

### Exercise: Homework Set

1. Verify  $(\{1, 2, 3, 4\}, \times \bmod 5)$  is a group by writing out its full multiplication table (Section 3 exercise, above).
2. Is  $3\mathbb{Z} = \{\dots, -6, -3, 0, 3, 6, \dots\}$  a subgroup of  $(\mathbb{Z}, +)$ ? Justify using the subgroup test (Section 3.1). Is  $\{0, 3, 6, 9, \dots\}$  (only the non-negative multiples of 3) a subgroup of  $(\mathbb{Z}, +)$ ? Why or why not?
3. List all distinct cosets of  $4\mathbb{Z}$  in  $\mathbb{Z}$ , confirm there are exactly 4, and identify which coset  $-5$  belongs to (Section 3.2 exercise, above).
4. Find a pair of zero divisors in  $\mathbb{Z}_{12}$ , and use the zero-divisor fact (Section 4.1) to explain why  $\mathbb{Z}_{12}$  cannot be a field, without computing any inverses.
5. Compute  $23 \times 15 \bmod 7$ ; list the non-invertible elements of  $\mathbb{Z}_8$ ; use the Extended Euclidean Algorithm (Section 6.1) to find the inverse of  $3 \bmod 7$  — confirm it matches the worked example.
6. Use the Extended Euclidean Algorithm to find  $5^{-1} \bmod 11$  (Section 6.1 exercise, above), showing all steps.
7. Is  $\mathbb{Z}_9$  a field? Why or why not? Answer it two ways: (a) using gcd (Section 6), and (b) using zero divisors (Section 4.1).
8. Show that  $\{(1, 1), (1, -1)\}$  is a valid basis of  $\mathbb{R}^2$  by (a) checking linear independence and (b) writing the vector  $(4, 2)$  as a linear combination of them.
9. Run Gram–Schmidt on  $b_1 = (1, 1)$ ,  $b_2 = (0, 2)$  (Section 7.2 exercise, above), and verify the result is orthogonal.
10. For the lattice generated by  $b_1 = (1, 2)$  and  $b_2 = (3, 1)$ : (a) is the point  $(7, 5)$  in the lattice? (b) compute  $\det(L)$  from this basis; (c) pick a different valid basis for the same lattice and confirm  $\det(L)$  is unchanged (Section 8.4 exercise, above).
11. Let  $L = \{(x, y) \in \mathbb{Z}^2 : x + y \text{ is even}\}$ . (a) Show  $L$  is a subgroup of  $(\mathbb{R}^2, +)$ . (b) Show  $L$  is discrete (find an  $\varepsilon$  that isolates  $\vec{0}$ ). (c) Since  $L$  is therefore a lattice, find a basis  $\{b_1, b_2\}$  for it, and compute  $\det(L)$ .
12. In your own words (3–4 sentences): why does the definition of a lattice need *both* “subgroup” and “discrete”? Use the  $\mathbb{Q}$  example from Section 8.2 to explain what goes wrong if you drop the discreteness requirement.
13. In your own words (3–4 sentences), explain why a “good basis” and a “bad basis” for the same lattice can make the same mathematical object feel completely different to work with computationally, now that you can measure this with norms (Section 7.1).